

1. SYSTEM ARCHITECTURE & OVERVIEW

GDRM (Gaurav Digital Rights Management) is an enterprise-grade, zero-trust document security ecosystem built with Flutter and Firebase. It provides end-to-end cryptographic encapsulation of PDF documents into proprietary .gdrm containers, offering hardware binding, username recipient locking, anti-copy reader protection, multi-factor destruction triggers, and watermarked physical paper printing.

2. FIREBASE AUTHENTICATION & RECIPIENT REGISTRY

User authentication is orchestrated through Firebase Auth and Cloud Firestore:

- Registration: Collects Full Name, Unique Preferred Username Handle (e.g., @gaurav), Email Address, and Password.
- Firestore Registry: User documents are indexed under gdrm_users/{username}, storing name, email, uid, and timestamps for real-time recipient verification.
- Flexible Login: Supports signing in via either unique username handle or email address with session persistence.
- Sidebar Identity Badge: Displays verified profile status, active username, and instant account switching.

3. USERNAME-LOCKED .GDRM CONTAINER ENCRYPTION

Senders can lock documents exclusively to a recipient username:

- Packer Verification: Sender inputs recipient username (e.g., @alice) with live Firestore registry lookup to confirm recipient existence.
- Cryptographic Binding: TARGET_USERNAME is embedded into the container header, and payload is encrypted using pseudo-random derived XOR keystreams.
- Unintended Receiver Enforcement: When a non-intended user opens the file, the app immediately blocks access and displays the Unintended Receiver Warning alert. No unauthorized decryption or prompt-leaking is permitted.

4. READER SECURITY SUITE & ANTI-COPY PROTECTION

- Zero Text Selection: Text selection, cursor drag highlighting, context copy menus, scroll status, and pagination dialogs are completely disabled in SfPdfViewer.
- Keyboard Interceptor: Intercepts and suppresses Ctrl+C, Cmd+C, Ctrl+A, Ctrl+X, and Ctrl+Insert to prevent clipboard leakage.
- Hardware Key Binding: Automatically locks file to the first opening device hardware key; sharing the file to an unauthorized machine trips the Piracy Alarm.
- FileRip Password Layer: Implements hashed password challenge with max attempt threshold before triggering irreversible file self-destruction.
- ChronoLock & Rigged Timers: Enables scheduled future unlock datetimes, live countdown time-bombs (active screen closes & file data melts on expiry), and max view count limits.

5. WATERMARKED PHYSICAL PAPER PRINTING ENGINE

PrintingService enforces paper-only distribution with persistent provenance tracking:

- Virtual PDF Printer Blocking: Inspects system spoolers and filters out virtual PDF writers (e.g., Microsoft Print to PDF, Adobe PDF, Save as PDF, Foxit, CutePDF, OneNote).
- Embedded Watermarking: Every page is rasterized and composited with high-resolution watermark metadata (Copyright Owner, Licensed To, Recipient Handle, Printing User, IP/MAC address, and Timestamp).
- Direct Hardware Spooling: Dispatches watermarked pages directly to the selected hardware printer.

6. .GDRM CONTAINER SPECIFICATION MATRIX

Field Header	Security Purpose & Functionality
LICENSED_TO	Authorized licensee/recipient display name
COPYRIGHT_OWNER	Legal copyright holder entity
FINGERPRINT	Unique cryptographic identifier: GDRM-XXXX-XXXX-XXXX
TARGET_USERNAME	Firebase recipient handle required for decryption
SENDER / RECEIVER	Cryptographic sender seed / bound device hardware key
OPEN_AT / RIGGED_EXPIRY	ChronoLock unlock datetime & time-bomb self-destruct UTC
MAX_OPENS / CURRENT_OPENS	Viewing ceiling counter threshold limit
TRAIL_START / TRAIL_END	Immutable SnailTrail forensic audit log timestamps